Squatting Kinematics and Kinetics and Their Application to Exercise Performance

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Abstract & Overview

Source: National Strength and Conditioning Association (NSCA) Guidelines & Schoenfeld Biomechanical Analysis (2010)
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1. Biomechanics of the Squat

The squat is a multi-joint compound movement involving simultaneous flexion and extension of the hips, knees, and ankles.
Key joint angles during the movement:
- Standing / Top Phase: Knee angle is approximately 170° to 180° (full extension without joint hyperextension). Torso angle is upright (near 180° relative to vertical).
- Parallel Depth: Thighs are parallel to the floor when the crease of the hip drops level with or below the top of the patella. This corresponds to a tibiofemoral (knee) angle of ≤ 100° (typically 80° to 95° for full parallel depth).
- Sub-Parallel Depth: Deep squats reach knee angles of 70° to 80°.

2. Common Form Errors

• Error A: Insufficient Depth ("TOO HIGH")
  - Biomechanical Cause: User reverses direction before knee angle reaches 100°.
  - Health / Muscle Impact: Limits quadriceps and gluteus maximus recruitment, shifting disproportionate shear forces to the patellar tendon without full posterior chain activation.
  - Correction: Focus on pushing the hips back and down, allowing knees to track forward over the toes while keeping heels firmly planted. Squat until thighs are at least parallel to the floor.
• Error B: Excessive Forward Torso Lean
  - Biomechanical Cause: Trunk angle drops below 130° relative to the hips/thighs.
  - Root Causes: Weak core stabilizers, limited ankle dorsiflexion, or quad fatigue leading to a "good morning" squat compensation.
  - Correction: Maintain a proud chest, actively brace the abdominal wall (Valsalva maneuver or diaphragmatic core engagement), and look forward at a fixed spot at eye level.
• Error C: Knee Valgus (Knees Caving Inward)
  - Biomechanical Cause: Adduction and internal rotation of the femur during the ascent phase.
  - Root Causes: Weak gluteus medius and hip abductors.
  - Correction: Cue "screw your feet into the floor" and push knees outward actively in line with the second toe throughout descent and ascent.

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3. Recommended Camera Setup for Squats

- Angle: 45-degree oblique or pure side (sagittal) view.
- Distance: 2.5 to 3.5 meters (8 to 11 feet) away.
- Framing: Full body framing must be maintained, ensuring both ankles and the head remain within the camera frame throughout the entire descent.